A3-PS001-TC171

Android Ransomware-like Activity Early Warning in a Controlled Lab

Institute of Engineering & Management, Kolkata

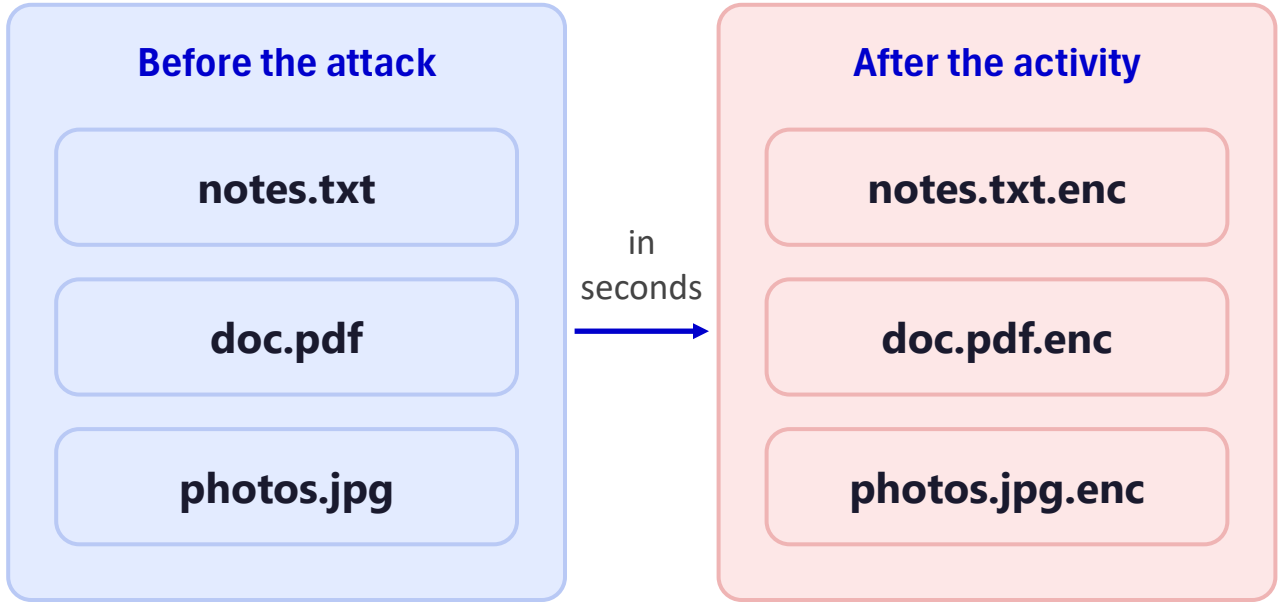
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Problem Description



Problem
Zero-day ransomware has no signature, but it renames many files within seconds.

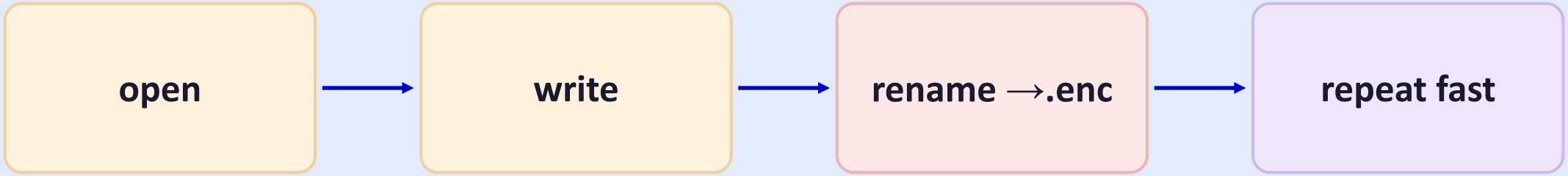
Approach
Hook file renames in a test directory; a rate rule flags bursts.

Outcome
3/3 simulated attacks detected, 0/2 false alarms (lab setting).

Goal: raise an early alert and stop only the test workload

Background

Ransomware file-system pattern



Android sandbox
An app cannot read another app's memory.

⇒

Our observation point
File renames inside one test directory.

Safe by design: emulator only • no real malware • no device-wide monitoring

Literature & RFC Review

Work	Approach	Key signals
Signature antivirus	Match known binaries	Hashes and patterns
HelDroid [3]	Static analysis of mobile ransomware	Lock-screen, encryption and threat-text features
CryptoLock [4]	Dynamic file-activity monitoring	Entropy, file-type change, similarity, deletion
ShieldFS [5]	Filesystem monitoring and recovery	I/O activity models, shadow copies
UNVEIL [6]	Automated dynamic analysis	File I/O traces, screen locking
This work	**Android hook on File.renameTo**	**Rename rate (≥ 10 in 2 s)**

Our focus: a lightweight, explainable rename-rate signal for Android. Entropy and other signals are future work.

No IETF RFC covers app-level file-activity monitoring, so Android security guidance [1] is used.

Proposed Solution

`Red_Team_Simulator.apk`
5 harmless workloads

→

`ueba_hook.js`
Frida hook on File.renameTo

→

`ueba_engine.py`
≥ 10 renames in 2 s ⇒ alert

→

`adb force-stop`
Test app only

Workloads in the APK (test directory)

CryptoLocker-style
simulated attack

DeepTree
simulated attack

PolyWiper
simulated attack

DocAutoSave
benign

PhotoSync
benign

Ethics and safety: no live malware • isolated emulator • only the test directory is touched

Optimization Proposed

Design decisions

Decision	Baseline	Ours
Trigger	Any rename	Renames in the test path
Decision	Single event	≥ 10 events in 2 s
Response	Manual stop	Auto force-stop of test app
Telemetry	None	Delay and files changed

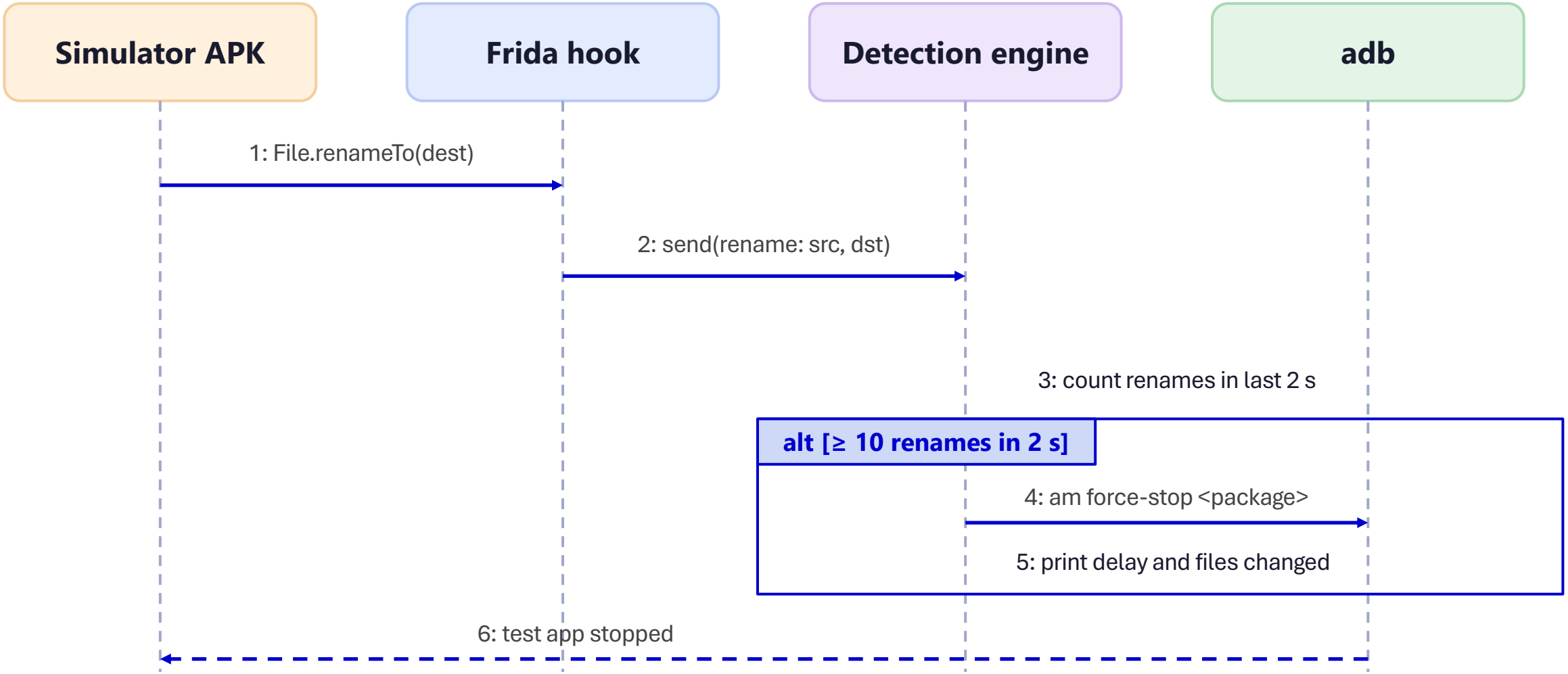
Lab results (5 runs)

Workload	Result	Delay	Files
CryptoLocker-style	Stopped	[s]	[n]
DeepTree	Stopped	[s]	[n]
PolyWiper	Stopped	[s]	[n]
DocAutoSave	Ignored	—	—
PhotoSync	Ignored	—	—

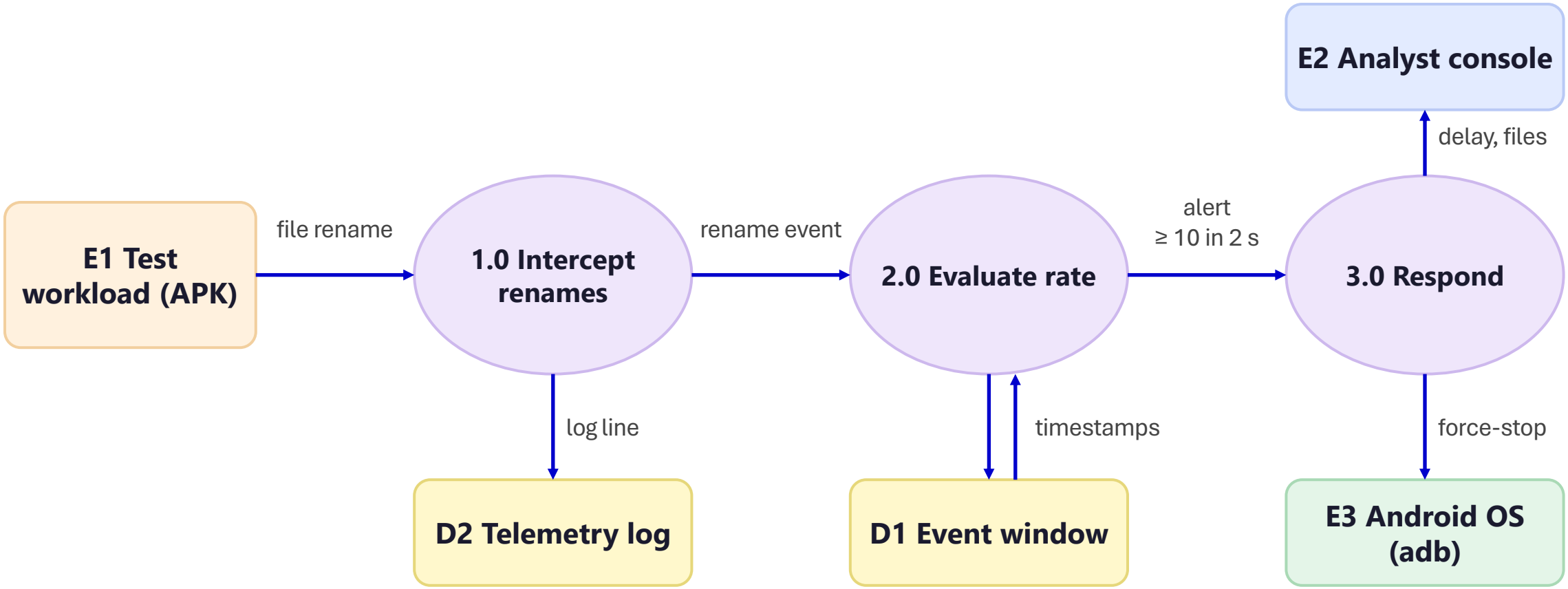
TP 3 • TN 2 • FP 0 • FN 0

Why 10 renames in 2 s? Benign workloads rename at most one file per 2 s, while attacks rename in bursts. 10 sits well above normal use. A threshold sweep is planned.

Solution Architecture & Design



Data Flow Diagram



Rectangle = external entity or data store • Circle = process • Arrow = data flow

Deployment Timeline

Week 1
Lab setup
Emulator, Frida

→

Week 2
Workloads
3 attacks, 2 benign

→

Week 3
Detector
Hook and engine

→

Week 4
Evaluate
Delay, files, repeats

→

Week 5
Demo & docs
Repo, PDF, demo

Known limitations (threat model)

Low and slow
Under 10 renames per 2 s is never flagged.

No rename
Overwrite, or delete-and-write, bypasses the rule.

Native code
libc rename() skips the Java File.renameTo hook.

Lab setup only
Frida and a host PC. Real use needs on-device monitoring.

Future work: libc-level hooks • file-entropy check • multi-signal score • on-device monitor

Bibliography

[1] Android Developers, "Security best practices," developer.android.com/privacy-and-security/security-best-practices (accessed Oct. 2026).

[2] Frida, "Frida: Dynamic instrumentation toolkit," frida.re/docs (accessed Oct. 2026).

[3] N. Andronio, S. Zanero, and F. Maggi, "HelDroid: Dissecting and detecting mobile ransomware," in Proc. RAID, 2015.

[4] N. Scaife, H. Carter, P. Traynor, and K. R. B. Butler, "CryptoLock (and drop it): Stopping ransomware attacks on user data," in Proc. IEEE ICDCS, 2016.

[5] A. Continella et al., "ShieldFS: A self-healing, ransomware-aware filesystem," in Proc. ACSAC, 2016.

[6] A. Kharraz et al., "UNVEIL: A large-scale, automated approach to detecting ransomware," in Proc. USENIX Security, 2016.



Thank You for Your Attention!

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